

→ Umbrella Term
AI →

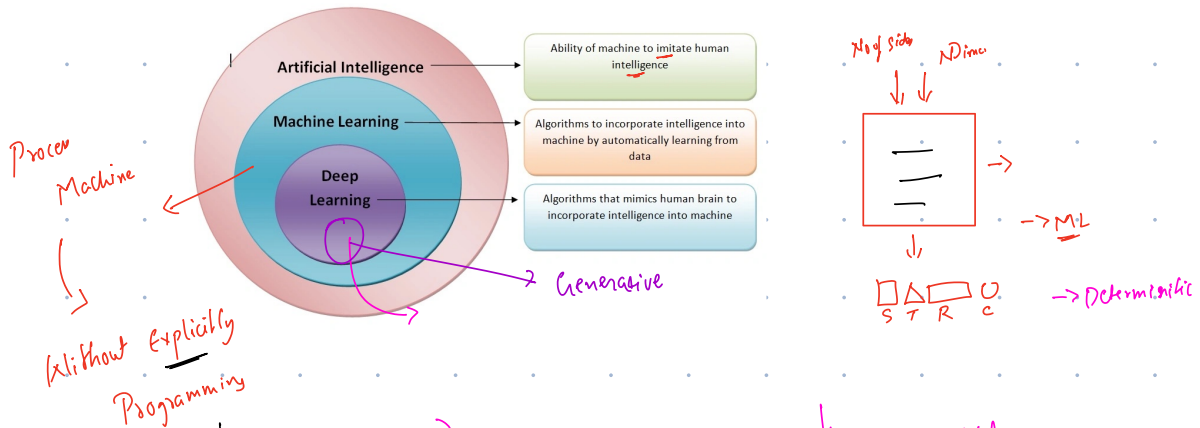
Branch - CS
(↳ Mimic - Human Intelligence

Inlet

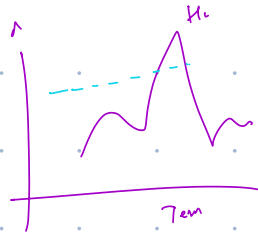
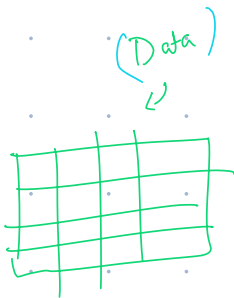
(Narrow) → Here - Mimic,

⇒ General - Equal Human

Strong - Beyond Human



↳ (Past Data) → (Trains) → (Patterns) → Predict
→ Probabilistic



↓
360 → Hist

□ □ Sq, Sq, F

- 0, 1, 2
- ① Supervised → Data + label
 - ② Unsupervised → Data
 - ③ Reinforcement → feedback
- ↳ Yes, No
-



iris Dataset
150 Rows
x 5 columns

- ① Setosa
- ② Versicolour
- ③ Virginica

Data (Independent)

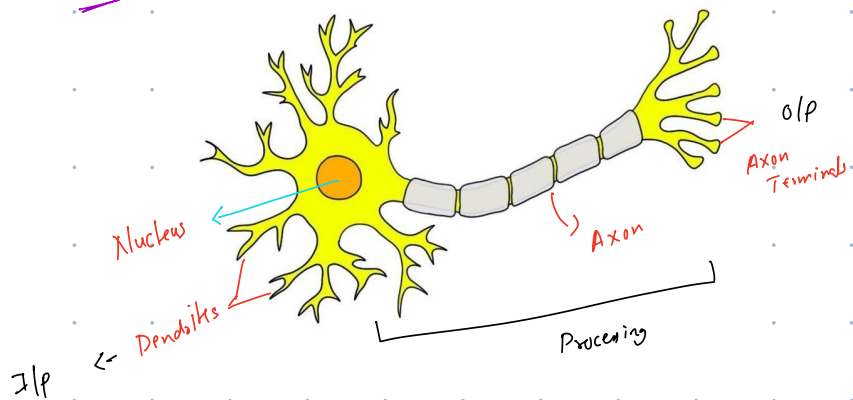
Depend

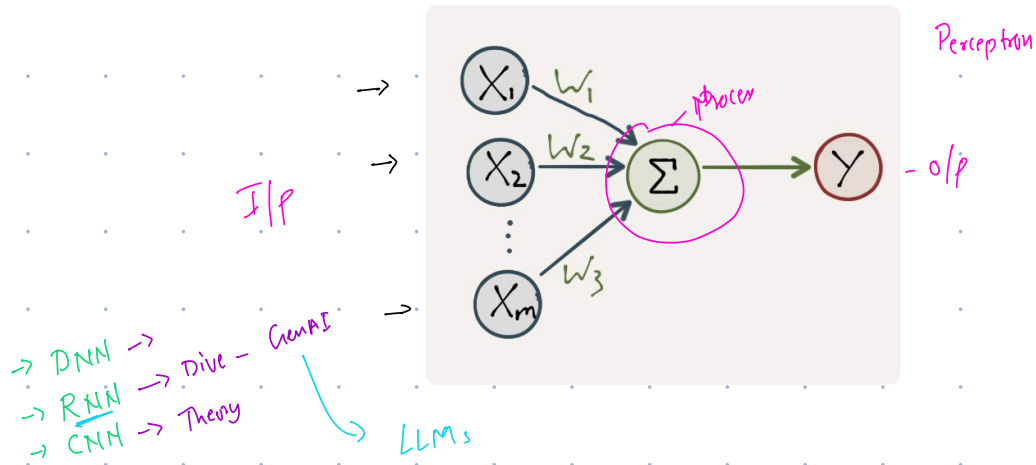
Target label

Pattern

	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9					setosa

Deep learning → Biological





GitHub

$\rightarrow$ VS Code / Colab
 $\rightarrow$ Python

① Python - Revision - ✓

② ML $\rightarrow$ Data Steps
 Supervised - Simple
 Unsuper - k-Mean

③ DL - ML Algorithm - ✓

ML - Explicitly - Data

lifecycle $\rightarrow$

[Iris.csv
SalaryData.csv]

① Data Mining $\rightarrow$ Gathering

② Data preprocessing

- Duplicate
 - Outliers
 - Missing
 - Datatypes $\rightarrow$ Age = 18, 19, 20
 inc

③

Dummy $\rightarrow$ mockup Data
 Survey
 Sensors
 CRM
 Repos $\rightarrow$ Kaggle
 GitHub

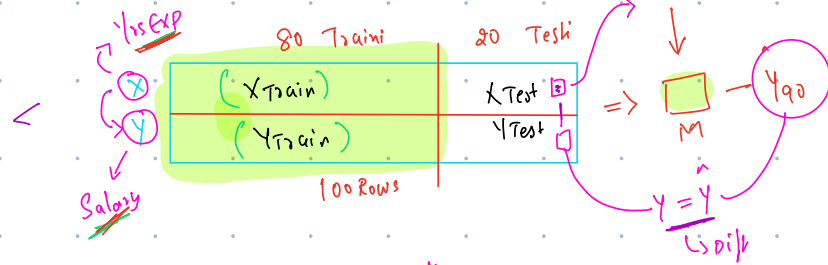
③ EDA

↳ Explore Data Analysis



④

Train Test Split



20 Test
15 - count
5 - wrong
75%

⑤

Regression

Classification

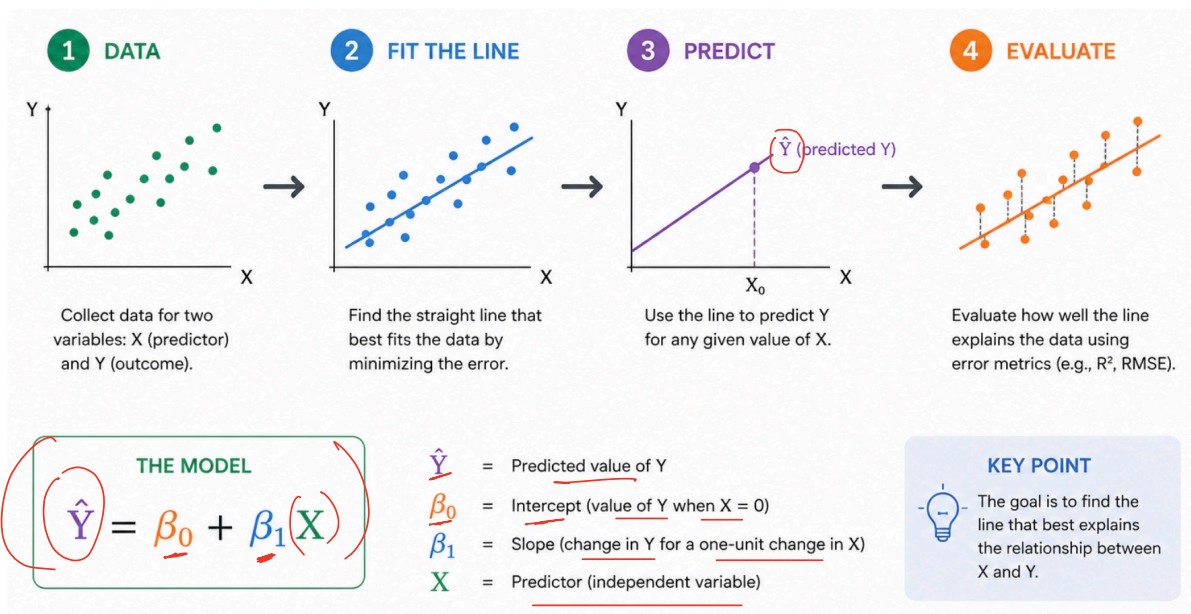
→ KNN - classifier

$X \rightarrow Y$ (One X , one Y)

⑥ Train

⑦ Test / Evaluation

⑧ Monitoring & Deployment



Handwritten notes:

$Y \rightarrow$ output

$\hat{Y} \rightarrow$ Predicted output

Data - 2 classes

A

B

Handwritten notes:

Supervised classification

$\leftarrow$ KNN - k Nearest Neighbors



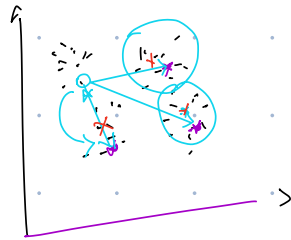
Unsupervised ML

↳ Data - labels

3 - Clusters $\leq$ $\begin{matrix} \text{Elite} \\ \text{Rich} \\ \text{Average} \end{matrix}$

People - Age, Money spent, Money earned, No of dependants, No of House. CIS

→ K-Means clustering



- 1) $K=3$ - clusters
- 2) Place the centroids
- 3) Euclidean distance Assign the points

Replace the centroids

$$10 + 28 + 3 / 3 = (-)$$

$$E = \text{Radius} = 3$$

$$\text{No of Radius points} = 3$$

